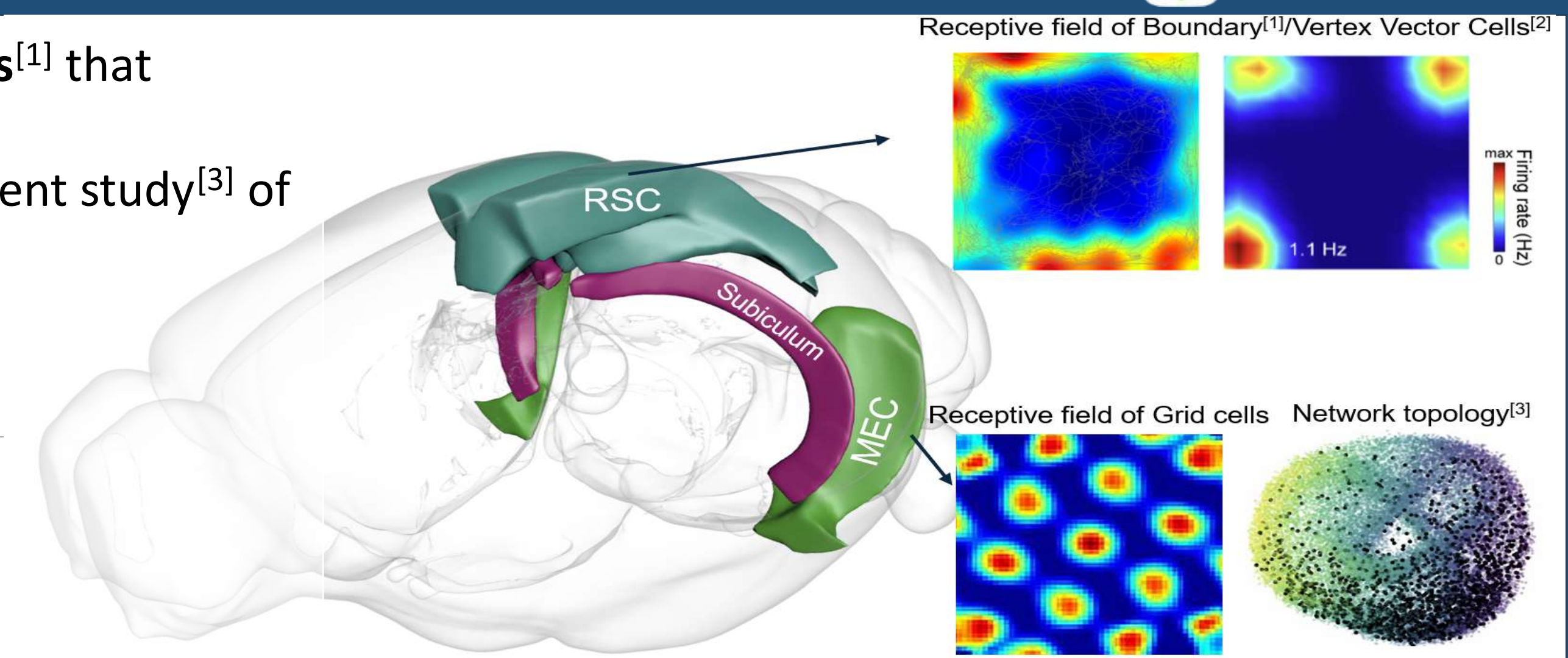


Introduction

- The Retrosplenial Cortex (RSC) has neurons representing environmental geometry such as **boundary vector cells**^[1] that respond to boundaries, or **vertex cells**^[2] that respond to vertices in the environment.
- Studying **network topology** can uncover broad mathematical characteristics of neural networks, such as the recent study^[3] of toroidal(donut) topology of grid cells in the medial entorhinal cortex.
- However, the network structure of boundary vector cells in the RSC is largely unknown. This study aims to leverage topological analysis as a tool to understand the following points:

Aims

- What is the topology of boundary vector cells in the RSC?
- How does topology change when RSC network is perturbed?
- What network structure underlies the boundary vector cell network?



Methods & Results

Methods

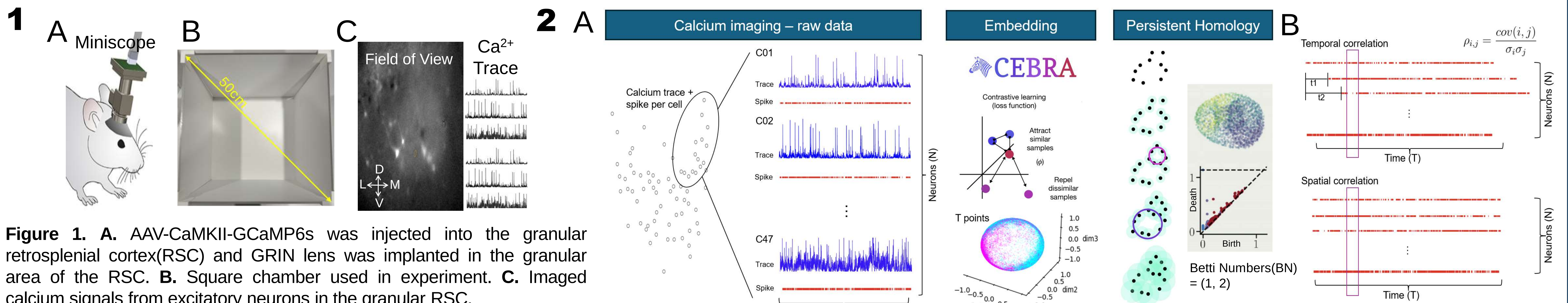


Figure 1. **A.** AAV-CaMKII-GCaMP6s was injected into the granular retrosplenial cortex(RSC) and GRIN lens was implanted in the granular area of the RSC. **B.** Square chamber used in experiment. **C.** Imaged calcium signals from excitatory neurons in the granular RSC.

Figure 2. **A.** Raw trace/spike data forms a neural matrix with ensemble data from each timestep forming each column(left). The neural matrix, after PCA whitening and denoising is embedded into lower dimensions to understand network topology. Unsupervised contrastive learning is used(CEBRA^[4], middle). Persistent homology is applied to understand topological shapes of high dimensional objects. Persistence diagrams plot the birth and death of topological objects(right). **B.** Temporal correlation was used to find ensembles from trace data with Pearson r values.

Result 1. Ensemble identification and ring topology of boundary vector cells

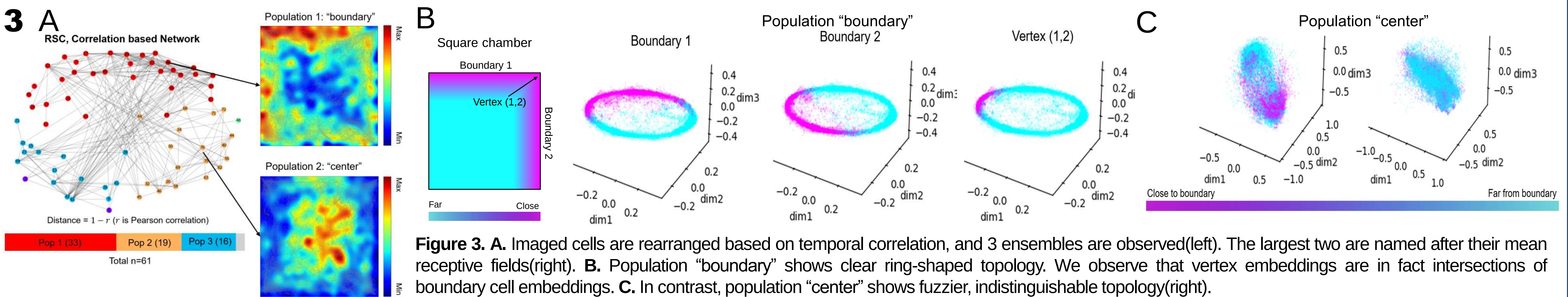


Figure 3. **A.** Imaged cells are rearranged based on temporal correlation, and 3 ensembles are observed(left). The largest two are named after their mean receptive fields(right). **B.** Population "boundary" shows clear ring-shaped topology. We observe that vertex embeddings are in fact intersections of boundary cell embeddings. **C.** In contrast, population "center" shows fuzzier, indistinguishable topology(right).

Result 2. Ring Topology of boundary vector cells transform into toroid topology with RSC network perturbation

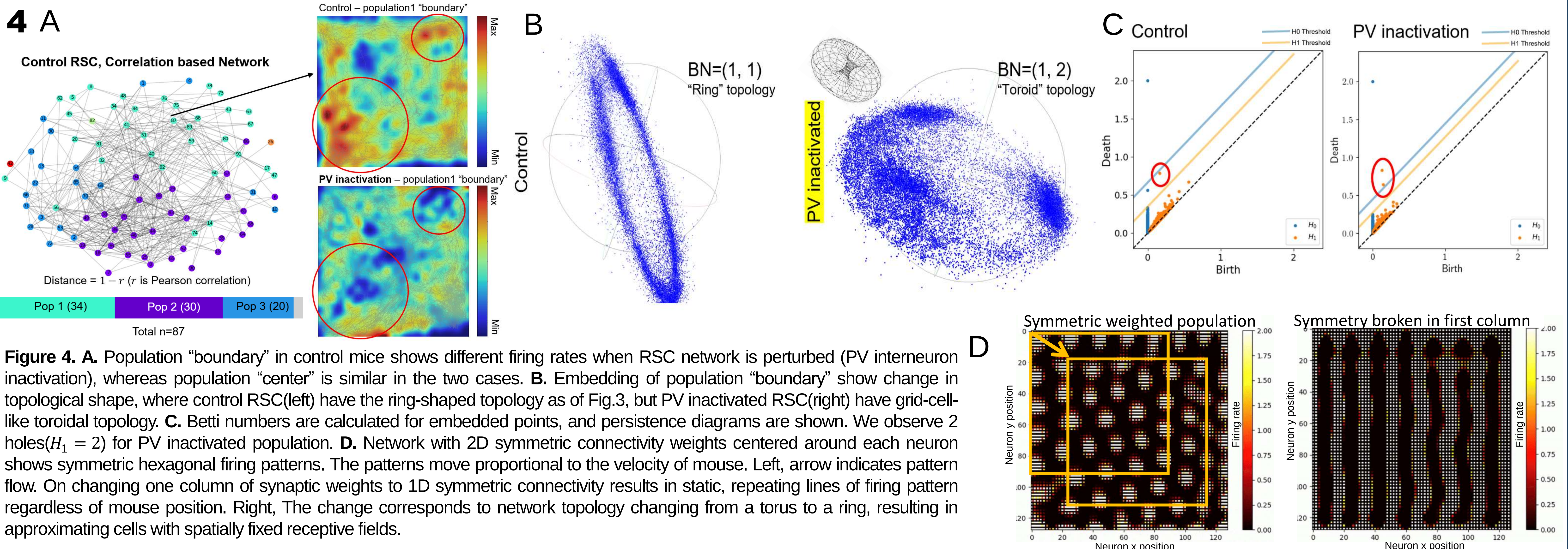


Figure 4. **A.** Population "boundary" in control mice shows different firing rates when RSC network is perturbed (PV interneuron inactivation), whereas population "center" is similar in the two cases. **B.** Embedding of population "boundary" show change in topological shape, where control RSC(left) have the ring-shaped topology as of Fig.3, but PV inactivated RSC(right) have grid-cell-like toroidal topology. **C.** Betti numbers are calculated for embedded points, and persistence diagrams are shown. We observe 2 holes($H_1 = 2$) for PV inactivated population. **D.** Network with 2D symmetric connectivity weights centered around each neuron shows symmetric hexagonal firing patterns. The patterns move proportional to the velocity of mouse. Left, arrow indicates pattern flow. On changing one column of synaptic weights to 1D symmetric connectivity results in static, repeating lines of firing pattern regardless of mouse position. Right, The change corresponds to network topology changing from a torus to a ring, resulting in approximating cells with spatially fixed receptive fields.

Discussion

- Ring-shaped topology was observed for boundary vector cells in the retrosplenial cortex.
- On perturbing the network (PV interneuron inactivation) in the RSC, the network topology of boundary vector cells transformed to a toroid, similar to the topology of a grid cell.
- Numerical simulations were done that could suggest that PV interneurons play an important role in fine-tuning the network to form spatially fixed receptive fields such as boundary cells.

Reference

[1] Barry et al., 2006, Journal of Neuroscience [2] Park et al., 2023, BioRxiv [3] Gardener et al., 2022, Nature [4] Schneider et al., 2023, Nature

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